



**CNP P.H. Robinson to W.A. Parish
345-kV Hardening Project – ERCOT
Independent Review (EIR) Scope**

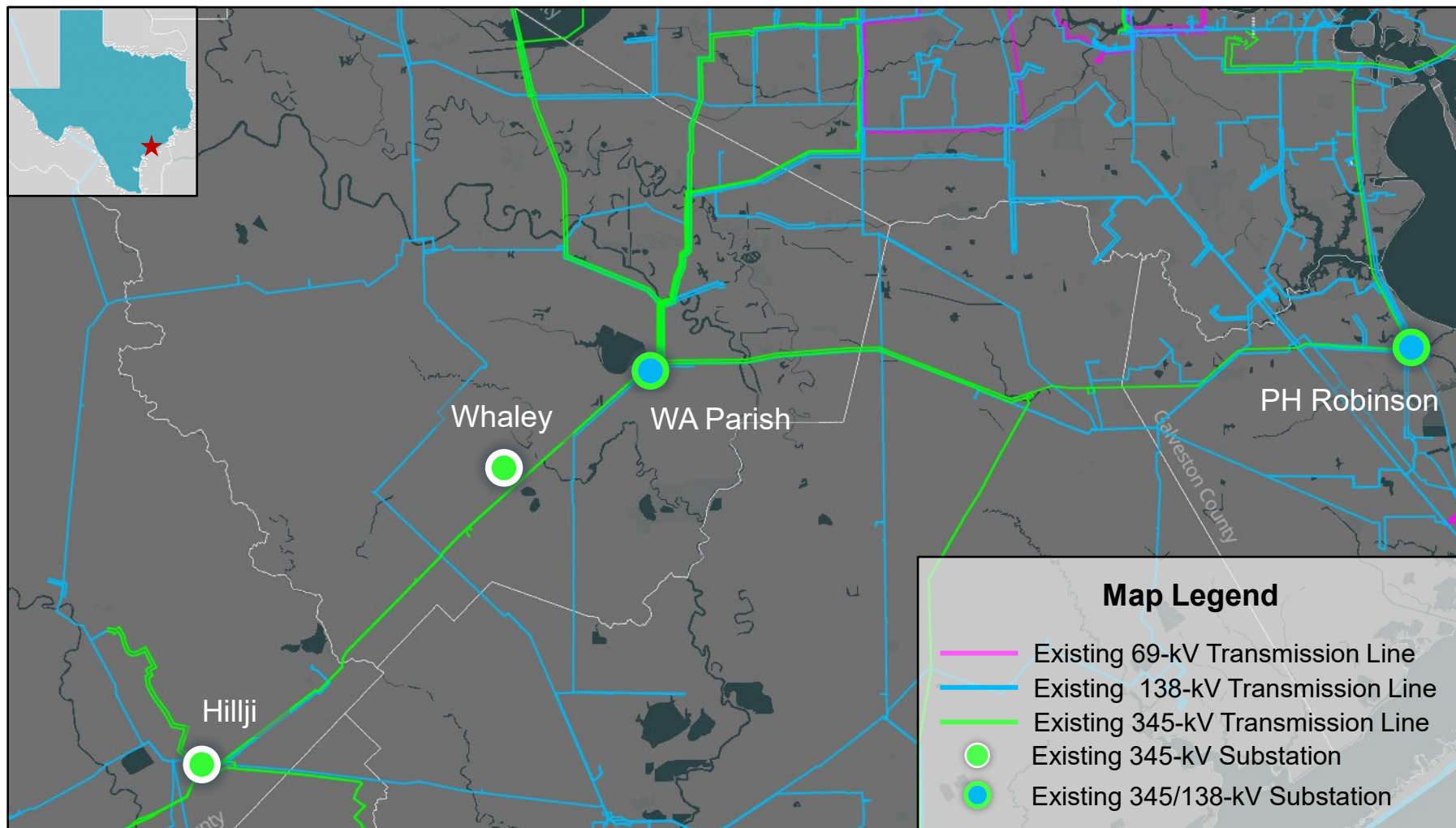
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RPG Meeting
September 25, 2025

Introduction

- CenterPoint Energy (CNP) submitted the P.H. Robinson to W.A. Parish 345-kV Hardening Project (25RPG028) for Regional Planning Group (RPG) review in August 2025
 - This Tier 1 project is estimated cost to cost \$511.3 million and will not require a Certificate of Convenience and Necessity (CCN)
 - Estimated in-service date (ISD) is December 2028
 - This project is needed to address aging infrastructure and to ensure resilience against hurricane-force winds and extreme cold conditions in the Galveston County
- ERCOT is currently conducting an ERCOT Independent Review (EIR)

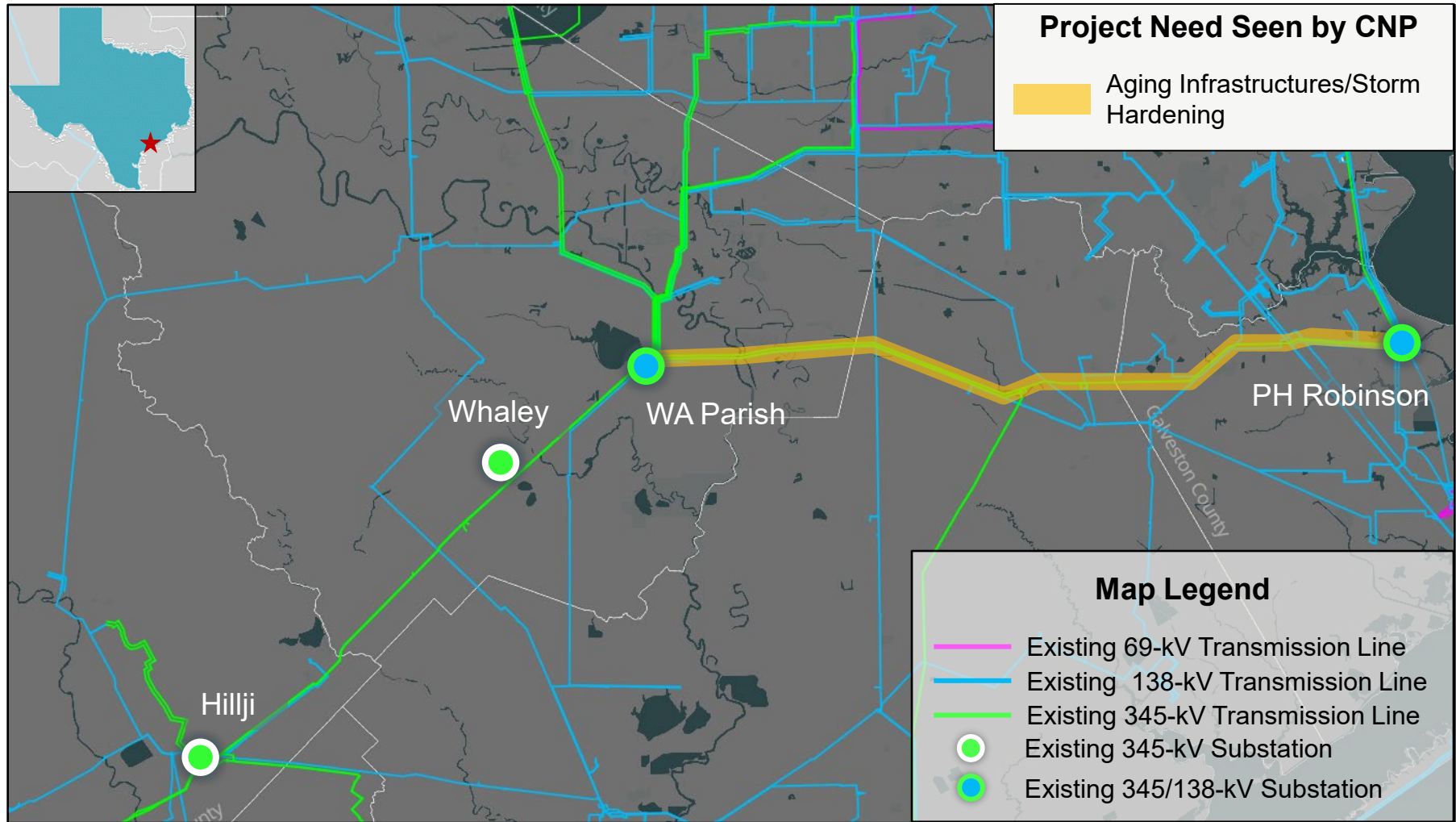
Study Area Map with Project Needs Seen by CNP



Proposed Project by CNP

- Rebuild/reconductor P.H. Robinson to W.A. Parish 345-kV double-circuit transmission line, with a normal and emergency ratings of at least 2,987 MVA, approximately 40.2-mile:
 - Rebuild/reconductor P.H. Robinson to Meadow Ckt.99 345-kV double-circuit transmission line, on double-circuit-capable steel structures with only one circuit in place, approximately 17.4-mile;
 - Ckt.93 will be rebuilt to the North of Ckt.99A within the existing transmission right of way (ROW). Ckt.99 will be rebuilt on double-circuit-capable lattice structures with conductor on both sides of the towers. One side will be Ckt.99, while the other side will be an energized tap;
 - Rebuild Meadow to Oasis Ckt.99 345-kV double-circuit transmission line, on double-circuit-capable steel structures with only one circuit in place, with a normal and emergency ratings of at least 2987 MVA, approximately 3.18-mile;
 - Rebuild Oasis to W.A. Parish Ckt.99 and Oasis to W.A. Parish Ckt.18 345-kV double-circuit transmission line on double-circuit-capable steel structures with both circuit in place, approximately 19.6-mile;
 - Upgrade Oasis 345-kV substation to 63 kAIC fault duty breaker rating;
 - Upgrade substation equipment to 5,000A;

Map with Proposed Project by CNP



Study Assumptions – Base Case

- Study Area
 - Galveston County in Coast Weather Zone is considered for this study
 - Brazoria and Harris Counties which are electrically close to project location
- Steady-State Base Case
 - Final 2024 Regional Transmission Planning (RTP) 2029 summer peak load case published on Market Information System (MIS) in December 2024, will be updated to construct the study base case
 - Case: 2024RTP_2029_SUM_12222024
 - Link: <https://mis.ercot.com/secure/data-products/grid/regional-planning>

Study Assumptions – Transmission, Load, & Reserve

- Based on the June 2025 Transmission Project and Information Tracking (TPIT) posted on the ERCOT website projects within the study area with in-service dates before December 2028 within the study area will be added to the study base case if not already modeled in the case
 - TPIT Link: <https://www.ercot.com/gridinfo/planning>
- Loads in study area
 - Load level in the study area will be maintained consistent with the Final RTP cases
- Reserve
 - The reserve will be maintained consistent with the 2024 RTP

Study Assumptions – Generation

- New generation that met the Planning Guide Section 6.9(1) condition with Commercial Operation Date (COD) before December 2028 in the study area at the time of the study will be added to the case if not already modeled in the RTP case based on August 2025 Generator Interconnection Status (GIS) report published in MIS in September 2025
 - Link: <https://www.ercot.com/gridinfo/resource>
- All generation will be dispatched consistent with the 2024 RTP methodology
- All recent retired/indefinitely mothballed units will be reviewed and opened (turned off), if not already reflected in the 2024 RTP Final cases

Contingencies & Criteria

- Contingencies

- NERC TPL-001-5.1 and ERCOT Planning Criteria
- (<http://www.ercot.com/mktrules/guides/planning/current>)
 - P0 (System Intact)
 - P1, P2-1, P7 (N-1 conditions)
 - P2-2, P2-3, P4, and P5 (345-kV only)
 - P3 (G-1+N-1: G-1 of Cedar Bayou, South Texas Project (STP), and WA Parish Unit 6)
 - P6-2 (X-1+N-1: X-1 of Cedar Bayou, Obrian, PH Robinson, and WA Parish 345/138-kV transformers)

- Criteria

- Monitor all 60-kV and above busses, transmission lines, and transformers in the study region (excluding generator step-up transformers)
- Thermal
 - Use Rate A for normal conditions
 - Use Rate B for emergency conditions
- Voltage
 - Voltages exceeding their pre-contingency and post-contingency limits
 - Voltage deviations exceeding 8% on non-radial load buses

Study Procedure

- Need Analysis
 - The reliability analysis will be performed to identify the need to serve the projected area load using the study base case
- Project Evaluation
 - Project alternatives will be tested to satisfy the NERC and ERCOT reliability requirements
 - ERCOT may also perform the following studies
 - Planned maintenance outage
 - Long-Term Load-Serving Capability Assessment
 - TSP(s) will provide Cost and Feasibility Assessment
- Generation and Load Scaling Sensitivity Analyses
 - Planning Guide Section 3.1.3(4)
- Subsynchronous Resonance (SSR) Assessment
 - Nodal Protocol Section 3.22.1.3(2)
- Congestion Analysis
 - Congestion analysis may be performed based on the recommended transmission upgrades to ensure that the identified transmission upgrades do not result in new congestion within the study area

Next Steps and Tentative Timeline

- Tentative Timelines
 - Status updates at the future RPG meetings
 - Final recommendation – Q1 2026

Thank you!



Stakeholder comments also welcomed through:

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